

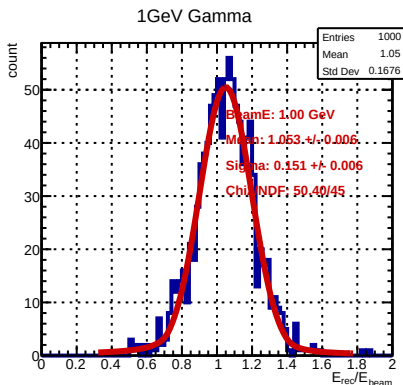
Different Kinds of Energy Regression

- Fitting Functions

- Linear function : $E_{\text{rec}} = p0 + p1 * E_{\text{ECAL}} + p2 * E_{\text{HCAL}}$
- Linear function w/ ratio : $E_{\text{rec}} = p0 + p1 * E_{\text{ECAL}} + p2 * E_{\text{HCAL}} + p3 * E_{\text{HCAL}} / (E_{\text{ECAL}} + E_{\text{HCAL}})$
- Quadratic function : $E_{\text{rec}} = p0 + p1 * E_{\text{ECAL}}^2 + p2 * E_{\text{HCAL}}^2$

- Weighting (Events enter fitting with a certain weight)

- Equal weighting : weighting = 1 (all the events are equal)
- Square-root of beam energy : weighting = $\sqrt{E_{\text{beam}}}$ (higher energy beam has more weight)



Linear energy regression

$$E_{\text{rec}} = p0 + p1 * E_{\text{ECAL}} + p2 * E_{\text{HCAL}}$$

Fit E_{rec} with double-side crystal ball function

- Energy resolution = σ / E_{beam} (%)
- Energy bias = μ / E_{beam}